Lit roads, lighter crashes? Street lighting, motorcycles and crash severity in accident records

Group name	Dark Horses
Dataset	kaggle.com/datasets/deeplumiere/accident-analysis-dataset
Code	github.com/.../cse558-midterm (Python scripts in `code/`, `README.md`, every reported number in `results/`)

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1. Dataset

Source: Kaggle: Downloaded CSV dataset on 25th October 2026. The dataset contains records of road accidents that took place in the UK between 2019 and 2025. Each row is one accident, and the columns describe additional information about the conditions at the time of accident.

What is raw about it: the accident ID was turned into scientific notation (`2.01E+07`) for 239,884 rows, which creates 239,659 false duplicate IDs; for every readable 2007–2010 ID the date is exactly 12 years later than the ID year; six district names lost the letters "na" (`Stevege` for Stevenage); 19,739 cells are missing; and Fatal is only 1.3% of rows.

	Raw file	After encoding
Instances (rows)	670,679	670,660
Attributes (columns)	14	52
Instances × attributes	9,389,506	34,874,320

Size check: $670{,}660 \times 52 = 34{,}874{,}320 \geq 3{,}000{,}000$, so the requirement is met (the raw file alone gives $670{,}679 \times 14 = 9{,}389{,}506$). The 52 columns are 4 numeric, severity as an ordinal, the KSI flag, district frequency, and 45 one-hot columns: light 5, road surface 6, road type 6, urban/rural 3, weather 9, vehicle type 16.

Key features

Feature	Type	Real-world meaning
`Accident_Severity`	ordinal	Slight, Serious or Fatal.
`Light_Conditions`	nominal	daylight, or darkness with lights lit, unlit, none or unknown
`Urban_or_Rural_Area`	nominal	urban or rural location
`Road_Type`	nominal	single or dual carriageway, roundabout, one-way, slip road
`Road_Surface_Conditions`	nominal	dry, wet, frost/ice, snow, flood
`Vehicle_Type`	nominal	vehicle recorded for the crash, 15 types
`Number_of_Casualties`	discrete	people injured, 1 to 68
`Number_of_Vehicles`	discrete	vehicles involved, 1 to 32
`District Area`	nominal	local-authority district, 416 levels after cleaning

2. Data preparation

Issue found	What we did	Why, and not the alternative
19 exact duplicate rows	dropped	the same crash counted twice would add weight to it in every test
ID in scientific notation for 239,884 rows	kept the ID out of every key and every test	de-duplicating on the ID would have deleted 239,659 distinct crashes
dates 12 years ahead of the ID year (all 420,776 readable older IDs)	date not used, and dropped from the encoding	no test needs time; shifting back would require guessing for the IDs that are unreadable
6 district names missing "na" (3,438 rows)	merged back: like `Stevege` → Stevenage	Test 2 pairs crashes within districts; a split district halves its counts and adds a fake one
missing categories: weather 14,351, road type 4,580, surface 742, urban/rural 26, vehicle 7	recoded to an explicit `Unknown` level	filling with the mode would push 4,580 unknown road types into "single carriageway", which is PS1's filter; `Unknown` rows simply fall out of each filter
51 missing coordinates	kept as missing	coordinates are not used
casualty outliers: IQR fence is $Q_3 + 1.5\,\mathrm{IQR} = 1$	kept all values (max 68)	a fence at 1 would flag all 158,994 multi-casualty crashes; they are real (e.g. bus crashes). Test 4 is re-run without the 67 crashes above 10 casualties
vehicles above fence 3.5 (15,310)	kept	real pile-ups; PS2 fixes `Number_of_Vehicles` $= 2$ anyway
6 nominal columns	one-hot (45 columns); severity ordinal $0/1/2$ plus a KSI flag	categories have no order; KSI is the response
district, 416 levels	frequency-encoded	one-hot would add 415 sparse columns; the raw label is kept for Test 2

3. What changed after preparation

Measure	Before	After
Rows	670,679	670,660
Attributes	14	52 (encoded)
Missing cells	19,739	0 in the columns we use; 51 coordinates left, unused
Duplicate rows	19	0
Duplicate IDs	239,659	ID no longer used as a key
District levels	422	416 (6 typo levels merged)
Usable dates	0 (shifted)	dropped
KSI share	14.67%	14.67% (unchanged)

Turning severity into the binary KSI (Killed or Seriously Injured) flag is what makes the proportion tests possible. Merging the district typos moved 3,438 crashes back to their real district, which changes the district-level pairs in Test 2. Only 19 rows were lost, so no group we test was thinned out by cleaning.

4. Basic questions and problem statement

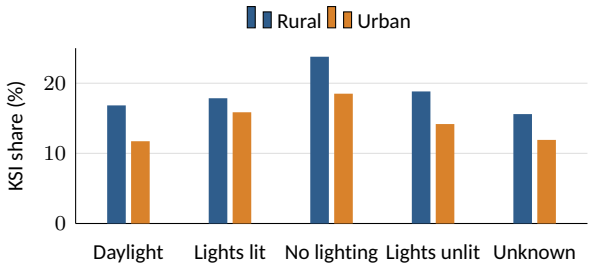


Figure 1: KSI share by light condition (all except Daylight are in darkness). Rural crashes on unlit roads are the most severe group: 23.8% KSI against 17.9% where lights are lit.

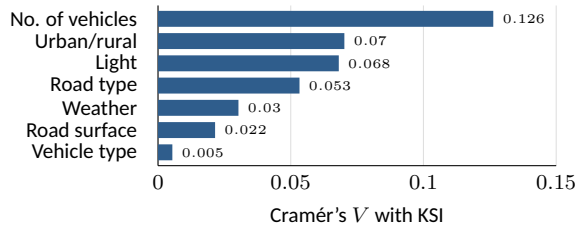


Figure 2: How strongly each column is associated with KSI. Vehicle type is the weakest by far ($V = 0.005$), 13 times weaker than light conditions. This warns us before PS2 is tested.

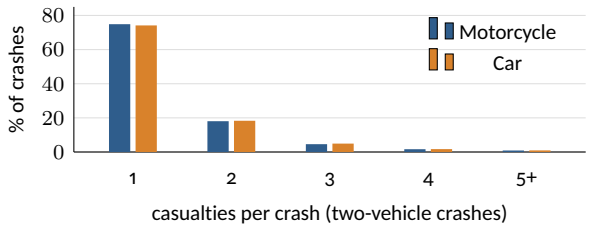


Figure 3: Casualties per crash are almost identical for motorcycles and cars: about 74% of crashes hurt exactly one person, and the counts are heavily right-skewed (skewness 7.0 over all crashes).

Basic questions

1. Which vehicle type is in the most crashes? Cars and taxis, in 77.4% of them.
2. Are rural crashes more severe than urban ones? Yes: 18.0% KSI against 12.8%.
3. Are crashes in the dark more severe than in daylight? Yes, in both areas (Figure 1).
4. Does bad weather make crashes worse? No: snowfall (9.7%) and icy roads (11.9%) have *lower* KSI shares than dry roads (15.1%), probably because people drive slower.

Our problem statements

PS1 (primary). Among rural single-carriageway crashes in darkness, is the KSI share higher where there is no street lighting than where lights are lit? Councils pay to light rural roads; if unlit crashes are no more severe, that spend is hard to justify on severity grounds.

PS2. In two-vehicle crashes, are motorcycle crashes more often KSI, while hurting *fewer* people per crash, than car crashes? If so, budgets split by casualty counts would understate motorcycle safety.

How they are novel: the basic questions compare one column against severity. PS1 compares two *darkness* groups on the same road type, so darkness itself is held fixed and only the lighting differs; it then checks the result within districts. PS2 asks two questions about the same groups, one on severity and one on casualty count, which could point in opposite directions. A plain "which vehicle has the most crashes" cannot show that.

5. Statistical inferences

Rows tested, PS1: 170,583 rural single-carriageway crashes, of which 28,201 were in darkness with no lighting (N) and 14,382 in darkness with lights lit (L). Excluded: "lights unlit" (608) and "lighting unknown" (1,683), whose lighting status is unclear, and daylight.

Rows tested, PS2: 397,950 two-vehicle crashes, of which 34,017 are motorcycles of any engine size (M) and 307,722 are cars or taxis (C). Other vehicles are excluded.

Test 1 (PS1): is the KSI share higher on unlit rural roads at night?

a) Hypotheses, written before running the test

H_0	$p_N - p_L \leq 0$
H_1	$p_N - p_L > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test (pooled SE for z, unpooled SE for the CI)
Smallest difference that matters	2 percentage points

p_N, p_L: KSI share of crashes on unlit and lit roads. One-sided, because the claim is that no lighting is worse. 2 pp means 20 more KSI crashes per 1,000 rural night crashes. That is roughly where the cost of a lighting scheme becomes worth discussing.

b) Other tests we considered

Two independent groups and a binary outcome, with expected cell counts in the thousands, call for a large-sample test on the difference of two proportions.

Also considered	Why we did not use it
Chi-square, 2×2	identical evidence ($\chi^2 = 65.32 = z^2$, $p = 6.4 \times 10^{-16}$ two-sided) but two-sided only, and it gives no CI for the difference
Fisher's exact test	needed only for small expected counts; our smallest is 3,348
CMH test by road surface	valid and stricter, since wet nights could confound. Run as a robustness check (below)
Logistic regression	adjusts for many confounders at once, but it is a model rather than one of our tests; left as an extension

c) Result

Decision

We **reject** H_0 at the 5% level. The data support a higher KSI share among unlit than lit rural single-carriageway crashes at night.

Estimate and uncertainty

Estimate	$+3.50$ pp (24.46% unlit, 20.96% lit)
95% confidence interval	$+2.67$ to $+4.33$ pp
Test statistic (d.f.)	$z = 8.08$ (normal; χ^2 form has 1 d.f.)
p-value	3.2×10^{-16}

What this tells us

The gap is about **35 more KSI crashes per 1,000 recorded crashes** in the unlit group. It concerns the severity of an injury crash, not the chance of having a crash or the number of people killed or seriously injured. The odds ratio of 1.22 means 22% higher *odds*.

Does the difference matter?

Yes, by our chosen 2 pp threshold: the whole 95% CI, $+2.67$ to $+4.33$ pp, is above it. This is evidence of a very important association between lights and Severity of the accidents. Although it still does not conclusively show that installing lights would remove the gap; speed limits and road layout could explain part of it.

Check: comparing the same road surfaces

Cochran–Mantel–Haenszel (CMH) test. This compares lit and unlit crashes within each road-surface category and combines the evidence. Its $z = 9.61$ and $p = 3.7 \times 10^{-22}$ support an association even after accounting for the different surface mixes.

Mantel–Haenszel risk difference. This is a weighted average of the KSI-share gaps within the surface categories. The estimate of $+4.18$ pp shows that the gap remains positive when comparisons are made within road surfaces, rather than only across all crashes.

Test 2 (PS1): does the gap hold within districts?

Test 1 treats every crash as independent, but crashes in one district share roads, policing and reporting habits. Here each district is its own control: for every district with at least 20 crashes in *both* groups we take $d_i = \text{KSI}_N - \text{KSI}_L$.

a) Hypotheses, written before running the test

H_0	$\mu_d \leq 0$
H_1	$\mu_d > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	paired t-test on the district differences d_i
Smallest difference that matters	2 percentage points

b) Other tests we considered

Each district gives a pair of shares, so the test should use the within-district difference. Skewness is 0.10, and the Shapiro–Wilk normality check finds no strong departure from normality ($p = 0.72$). This supports using the t-test, but does not prove the differences are normal.

Also considered	Why we did not use it
Wilcoxon signed-rank	uses ranks of paired differences rather than their mean; assumes symmetry. Run alongside: $W = 20{,}981$, $p = 2.3 \times 10^{-7}$
Sign test	valid, but uses only the sign of each d_i, so it has less power: 158 of 247 positive, $p = 6.7 \times 10^{-6}$
Unpaired Welch t-test	ignores the pairing and leaves district-to-district variation in the error: $t = 4.80$, $p = 1.1 \times 10^{-6}$, weaker than the paired $t = 5.57$ because the two shares are correlated across districts ($r = 0.26$)

c) Result

Decision

We **reject** H_0 at the 5% level. The average within-district KSI-share difference is positive among the districts that meet our sample-size rule.

Estimate and uncertainty

Estimate	mean $d = +3.08$ pp (median $+2.93$ pp; 247 of 396 districts, 37,189 crashes)
95% confidence interval	$+1.99$ to $+4.16$ pp
Test statistic (d.f.)	$t = 5.57$ (246)
p-value	3.3×10^{-8}

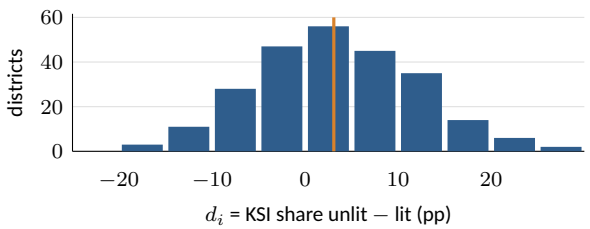


Figure 4: Test 2: KSI-share differences across 247 districts. The orange line marks the mean, $+3.08$ pp; the spread shows that the gap varies across districts.

What this tells us

Each district has equal weight here, unlike Test 1, where each crash has equal weight. The $+3.08$ pp average therefore describes a district-level difference, not another estimate of exactly the same pooled quantity.

Of the 247 districts, **158 (64%) have a positive gap** and 89 have a negative gap. The SD of 8.7 pp shows substantial variation: the overall finding is not true in every district.

Does the difference matter?

The estimate exceeds 2 pp, but the CI starts at 1.99 pp, just below the threshold. We have clear evidence of a positive average gap, but cannot confidently say the district-level gap exceeds our practical threshold.

Checks: is the finding tied to the t-test?

Sign test. This counts only whether each district's difference is positive or negative, ignoring its size. With 158 of 247 positive and $p = 6.7 \times 10^{-6}$, it supports a positive gap in more than half of the qualifying districts.

Pairing and scope. The paired test is more precise than the unpaired comparison here ($t = 5.57$ vs 4.80), consistent with the positive correlation between each district's two shares. These checks use the same data, not independent replications, and the conclusions exclude the 149 districts that do not meet the minimum counts.

Test 3 (PS2): are motorcycle crashes more often KSI?

a) Hypotheses, written before running the test

H_0	$p_M - p_C \leq 0$
H_1	$p_M - p_C > 0$
Significance level	$\alpha = 0.05$
Side	one-sided (right)
Test	two-proportion z-test
Smallest difference that matters	5 percentage points

p_M, p_C: KSI share of two-vehicle crashes recorded with a motorcycle or a car. A gap below 5 pp would not justify a separate motorcycle-safety budget line.

b) Other tests we considered

Again two independent groups and a binary outcome with large counts.

Also considered	Why we did not use it
Chi-square, severity (3 levels) × group	valid, but tests any difference in the severity mix, not our directional KSI question. Supporting run: $\chi^2 = 1.65$ (2 d.f.), $p = 0.44$
Fisher's exact test	counts are far too large to need it (smallest expected count 301)
Relative risk with CI	useful alongside, but our threshold is an absolute difference: RR $= 0.986$ (0.956 to 1.017)

c) Result

Decision

We **fail to reject** H_0 at the 5% level. This test does not support the claim that motorcycle-labelled crashes are more often KSI than car-labelled crashes. It does not prove that the two proportions are exactly equal.

Estimate and uncertainty

Estimate	-0.16 pp (11.36% motorcycle, 11.52% car)
95% confidence interval	-0.52 to $+0.19$ pp
Test statistic (d.f.)	$z = -0.89$ (normal)
p-value	0.81

What this tells us

The estimated motorcycle-minus-car difference is slightly negative, but its CI includes zero. Both a small decrease and a small increase are compatible with these data. We therefore cannot claim that either group has a higher KSI share from this comparison.

Does the difference matter?

The CI runs from -0.52 to $+0.19$ pp, far below the proposed $+5$ pp increase. Under the test assumptions, the data rule out an increase of that size: a precise negative finding for our proposed effect.

Checks: severity levels and relative risk

Three-level chi-square. This keeps Fatal, Serious and Slight separate, so it can find differences hidden by combining Fatal and Serious into KSI. Its $\chi^2 = 1.65$ (2 d.f.), $p = 0.44$, also gives no evidence of a different severity mix; expected counts are adequate.

Relative risk. This divides the motorcycle KSI proportion by the car proportion; a ratio of 1 means equal proportions. The estimate is 0.986 with a CI of 0.956 to 1.017, which includes 1 and agrees with the small absolute difference.

Test 4 (PS2): do motorcycle crashes hurt fewer people?

a) Hypotheses, written before running the test

$\boldsymbol{H_0}$	$\mu_M - \mu_C \geq 0$
$\boldsymbol{H_1}$	$\mu_M - \mu_C < 0$
Significance level	$\alpha = 0.05$
Side	one-sided (left)
Test	Welch t-test (unequal variances)
Smallest difference that matters	0.10 casualties per crash

μ_M, μ_C: mean casualties per two-vehicle crash. 0.10 means 10 fewer injured people per 100 crashes, enough to change ambulance and hospital load.

b) Other tests we considered

Two independent groups and a mean of counts. With n in the tens of thousands, the CLT makes the sample means normal despite the skew (Figure 3), and σ is unknown, so we use a two-sample t-test. We chose Welch's version because the spreads differ.

Also considered	Why we did not use it
Pooled t-test	assumes equal variances, rejected by Levene ($W = 11.0$, $p = 0.0009$) and by the F-test ($F = 0.93$, $p = 10^{-17}$). Run anyway: $t = -3.32$, $p = 0.0005$, same decision
Two-sample z-test	needs the population σ; at this n it gives the same $z = -3.41$, $p = 0.0003$
Mann–Whitney U	74% of values are tied at 1, and it tests a shift in distribution, not the mean. Robustness run: $p = 0.001$, $P(M > C) + \frac{1}{2}P(\text{tie}) = 0.496$, practically a coin flip
Poisson rate test	casualties start at 1, and their variance is *below* the mean (var/mean $= 0.44$ for M, 0.47 for C); shifted to start at 0 they are overdispersed (1.65, 1.70). Either way the Poisson variance is wrong; it gives $p = 0.012$

c) Result

Decision

We **reject** H_0 at the 5% level. Motorcycle-labelled two-vehicle crashes have a lower mean casualty count than car-labelled crashes in this file.

Estimate and uncertainty

Estimate	-0.015 casualties per crash (1.362 motorcycle, 1.377 car)
95% confidence interval	-0.024 to -0.006
Test statistic (d.f.)	$t = -3.41$ (42,493)
p-value	3.2×10^{-4}

What this tells us

The gap is about **1.5 fewer casualties per 100 recorded crashes**, with a CI of about 0.6 to 2.4 fewer. This is a difference in the total people injured per crash.

Does the difference matter?

Not by our chosen standard of 0.10 casualties per crash, or 10 per 100 crashes. The whole CI is much closer to zero than that threshold. With 341,739 crashes, a small difference can be statistically detectable without being practically important; rejecting H_0 is not enough on its own.

Checks: variance and distribution

Levene's test. This checks whether the groups have equal spreads. Its $p = 0.0009$ supports unequal variances, reinforcing the use of Welch rather than the equal-variance pooled test; the classical F-test is less reliable for these skewed counts.

High-casualty sensitivity check. Removing the 67 crashes with more than 10 casualties leaves a difference of -0.0146 and $p = 3.3 \times 10^{-4}$, close to the original result. The

small gap is therefore not driven only by the most extreme counts.

What do Tests 3 and 4 say together?

The proposed contrast in PS2 does not appear: motorcycle-labelled crashes show no evidence of a higher KSI share, and their lower casualty count is too small to meet our practical threshold. These are findings about recorded crash labels, not proof that motorcycles are safe.

Check: four primary tests at once

Holm correction controls the chance of at least one false rejection across the four primary tests at 5%, provided their p-values are valid. It makes the evidence threshold stricter without requiring the tests to be independent.

Test	Holm p-value	Decision
1	1.3×10^{-15}	reject H_0
2	9.8×10^{-8}	reject H_0
3	0.81	fail to reject H_0
4	6.5×10^{-4}	reject H_0

No primary decision changes after correction. The supporting checks help assess assumptions and interpretation, but use the same data and are not independent confirmations of the findings.

6. Limitations

- **No speed-limit column in our datatset**. In Great Britain a road with street lighting defaults to 30 mph, while an unlit rural single carriageway defaults to 60 mph. Thus, Part of the 3.5 pp gap in PS1 may be speed, not light, and PS2 cannot be adjusted for speed either.
- **Association, not cause.** Lights are not placed at random: they go on junctions and built-up stretches. Only a before-and-after study of newly lit roads could show cause.
- **`Vehicle_Type` looks detached from the crash.** Its near-zero association with every other column means PS2's conclusions describe this file, not motorcyclists in Great Britain.
- **Injury crashes only.** STATS19 records crashes with at least one injury that were reported to police, so the KSI share is a share of injury crashes, and slight injuries are under-reported.
- **Districts kept.** Test 2 uses 247 of 396 districts; the 149 with fewer than 20 crashes in a group are left out, and the result may not apply to them.
- **Shifted dates.** Without usable dates we cannot check seasonal confounding (dark evenings in winter coincide with wet roads); the CMH check by road surface covers only part of this.